



Insurance Premium — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

Version 1.0 — 04-April-2026

David K Kelly

CONFIDENTIAL — PROPRIETARY

Legal Notice

PROPRIETARY AND CONFIDENTIAL

This document, including all algorithms, models, methodology, formulas, calculations, and intellectual concepts contained herein, constitutes the exclusive intellectual property and confidential trade secrets of MKM Research Labs (“MKM”).

Any usage, reproduction, distribution, modification, or disclosure of this document or any portion thereof, without the express prior written authorisation from MKM Research Labs is strictly prohibited and constitutes an infringement of intellectual property rights.

The document is provided on an “as is” basis. MKM Research Labs makes no representations or warranties of any kind, express or implied, regarding its accuracy, reliability, or suitability for any particular purpose.

All rights reserved. © 2021–2026 MKM Research Labs.

Governed by the laws of the United Kingdom.

1 Test Results — Insurance Premium (MKM-IP-001)

Tests run on **04 April 2026 at 18:43:46**.

Table 1: Test Results Summary — Insurance Premium (MKM-IP-001)

Total Tests	36
Passed	36
Failed	0
Skipped	0
Duration	0.00s

Table 2: Detailed Test Results — Insurance Premium

Test Description	Acceptance Criteria	Result	Time
All factors applied	<code>abs(result - round(expected, 2)) < 0.01</code>	Pass	0.000s
Basic calculation	<code>result == 1250.0</code>	Pass	0.000s
Claims factor increases premium	<code>with_claims > base</code>	Pass	0.000s
Clamped to max	<code>result == MAX_PREMIUM</code>	Pass	0.000s
Clamped to min	<code>result == MIN_PREMIUM</code>	Pass	0.000s
Contents premium added	<code>with_contents == base + 500.0</code>	Pass	0.000s
High flood risk increases premium	<code>high_flood > low_flood</code>	Pass	0.000s
Returns float	<code>isinstance(result, float)</code>	Pass	0.000s
Security factor reduces premium	<code>with_security < without</code>	Pass	0.000s
100 years is heritage	<code>get_age_premium.band(1925, current_year=2025) == 'heritage'</code>	Pass	0.000s
10 years is modern	<code>get_age_premium.band(2015, current_year=2025) == 'modern'</code>	Pass	0.000s
30 years is established	<code>get_age_premium.band(1995, current_year=2025) == 'established'</code>	Pass	0.000s
Default current year	<code>result in AGE_PREMIUM_FACTORS</code>	Pass	0.000s
Established	<code>get_age_premium.band(1960, current_year=2025) == 'established'</code>	Pass	0.000s
Exactly 29 years modern	<code>get_age_premium.band(1996, current_year=2025) == 'modern'</code>	Pass	0.000s
Exactly 99 years established	<code>get_age_premium.band(1926, current_year=2025) == 'established'</code>	Pass	0.000s
Exactly 9 years new build	<code>get_age_premium.band(2016, current_year=2025) == 'new_build'</code>	Pass	0.000s
Heritage old	<code>get_age_premium.band(1850, current_year=2025) == 'heritage'</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Modern	<code>get_age_premium_band(2000, current_year=2025) == 'modern'</code>	Pass	0.000s
New build	<code>get_age_premium_band(2020, current_year=2025) == 'new.build'</code>	Pass	0.000s
Exactly 100 third band	<code>(lo, hi) == AREA_PREMIUM_BANDS[2][1]</code>	Pass	0.000s
Exactly 50 second band	<code>(lo, hi) == AREA_PREMIUM_BANDS[1][1]</code>	Pass	0.000s
Just below 100 second band	<code>(lo, hi) == AREA_PREMIUM_BANDS[1][1]</code>	Pass	0.000s
Just below 50 first band	<code>(lo, hi) == AREA_PREMIUM_BANDS[0][1]</code>	Pass	0.000s
Large area fourth band	<code>(lo, hi) == AREA_PREMIUM_BANDS[3][1]</code>	Pass	0.000s
Medium area second band	<code>(lo, hi) == AREA_PREMIUM_BANDS[1][1]</code>	Pass	0.001s
Returns tuple	<code>isinstance(result, tuple); len(result) == 2</code>	Pass	0.000s
Small area first band	<code>(lo, hi) == AREA_PREMIUM_BANDS[0][1]</code>	Pass	0.000s
Age premium factors all positive	<code>lo > 0; hi >= lo</code>	Pass	0.000s
Area bands are sorted	<code>thresholds == sorted(thresholds)</code>	Pass	0.000s
Base rate range	<code>lo > 0; hi > lo</code>	Pass	0.000s
Construction type factors present	<code>'Brick' in CONSTRUCTION_TYPE_FACTORS; 'Timber Frame' in CONSTRUCTION_TYPE_FACTORS</code>	Pass	0.000s
Flood risk factors increase with risk	<code>very_high_hi > very_low_lo</code>	Pass	0.000s
Flood risk factors present	<code>set(FLOOD_RISK_PREMIUM_FACTORS.keys()) == expected</code>	Pass	0.000s
Min max premium bounds	<code>MIN_PREMIUM == 200; MAX_PREMIUM == 20_000</code>	Pass	0.000s
Property type factors present	<code>set(PROPERTY_TYPE_PREMIUM_FACTORS.keys()) == expected</code>	Pass	0.000s

1.1 Test Document History

Date	Version	Description	Author
04-Apr-2026	1.0	Initial test suite (36 tests)	D. Kelly